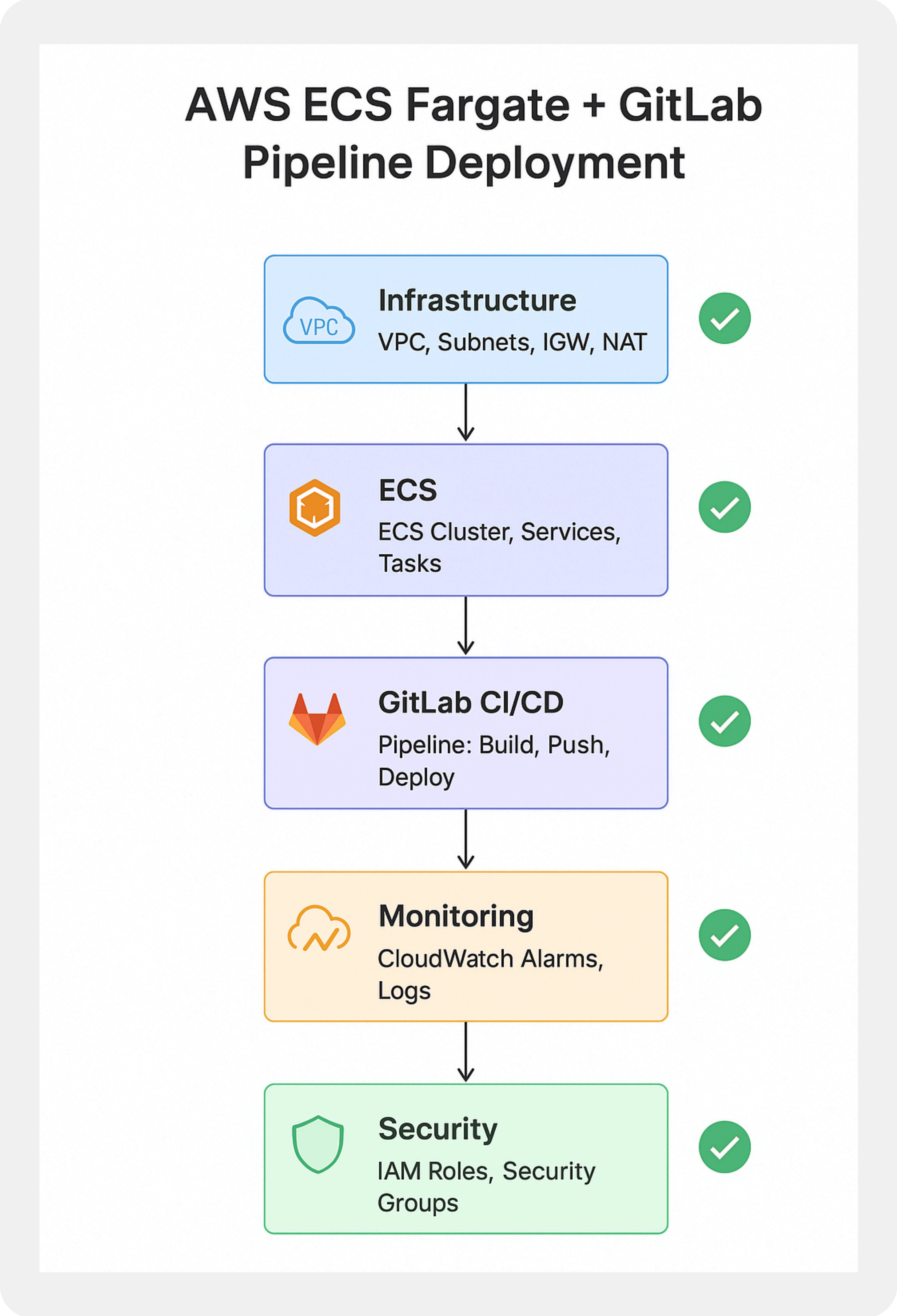
Highly Available Web Application Deployment on AWS with Full CI/CD and Monitoring

TASK & GOAL

Task : Deploy Employee Management REST API to AWS (ECS + Fargate) with full CI/CD, infra-as-code, monitoring, backups, and CDN

Build a repeatable, secure, observable pipeline that builds a containerized FastAPI Employee Management service, pushes the image to ECR, deploys to ECS (Fargate) via CloudFormation, serves traffic through CloudFront + Route53, stores backups in S3, and monitors via CloudWatch/CloudTrail. CI/CD runs in GitLab

FLOW


AWS ECS Fargate + GitLab Pipeline Deployment
Infrastructure
VPC
VPC, Subnets, IGW, NAT
ECS
ECS Cluster, Services, Tasks
GitLab CI/CD
Pipeline: Build, Push, Deploy
Monitoring
CloudWatch Alarms, Logs
Security
IAM Roles, Security Groups

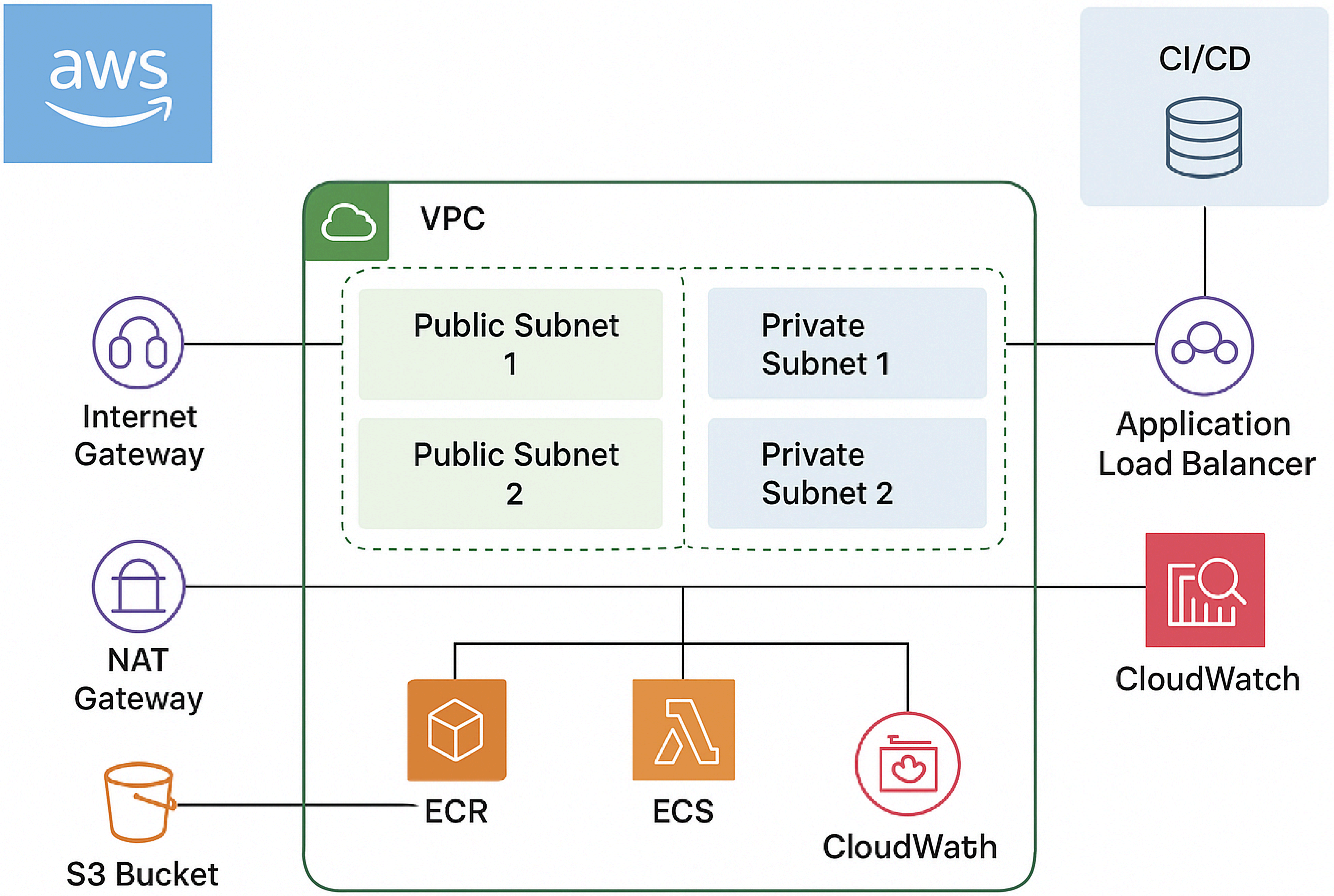
LOCAL DEPLOYMENT

- BUILD USING DOCKER
- DEPLOY USING DOCKER
- PUSH INTO GITLAB REPO

Build a application in local and push it into repo

CREATE INFRASTRUCTURE FOR THIS PROJECT:

- ✅ VPC
- ✅ Subnets (Public + Private)
- ✅ Internet Gateway & NAT Gateway
- ✅ Security Groups
- ✅ ECS Cluster & Service
- ✅ Task Definition
- ✅ ECR + S3
- ✅ ALB + Target Group
- ✅ IAM Roles
- ✅ CloudWatch + CloudTrail


aws
CI/CD
VPC
Public Subnet 1
Private Subnet 1
Public Subnet 2
Private Subnet 2
Internet Gateway
Application Load Balancer
NAT Gateway
CloudWatch
ECR
ECS
CloudWath
S3 Bucket

REPOSTORIES

GITLABPIPELINE DEPLOY IN FARGATE

1.CREATE A REPO FOR FILES

REPO: HTTPS://GITLAB-CICD.CODEWAVE.COM/AKHILBATTU/TASK

2.GITLAB PIPELINE:

INCLUDES : BUILD → PUSH TO ECR → DEPLOY

EXPOSE TO INTERNET

LOADBALANCERDNS :

HTTP://MYAPP-DE-ALB-EIHCSVP0LIZI-1.ELB.AMAZONAWS.COM/HEALTH

cloud front url

https://dykgxyb2pbth/health

For more detailds reffer this document

https://docs.google.com/document/d/1FMv5idcoXMSbyGv06V-v_TyRsiHBPqV-zIcngT9JCtc/edit?tab=t.0